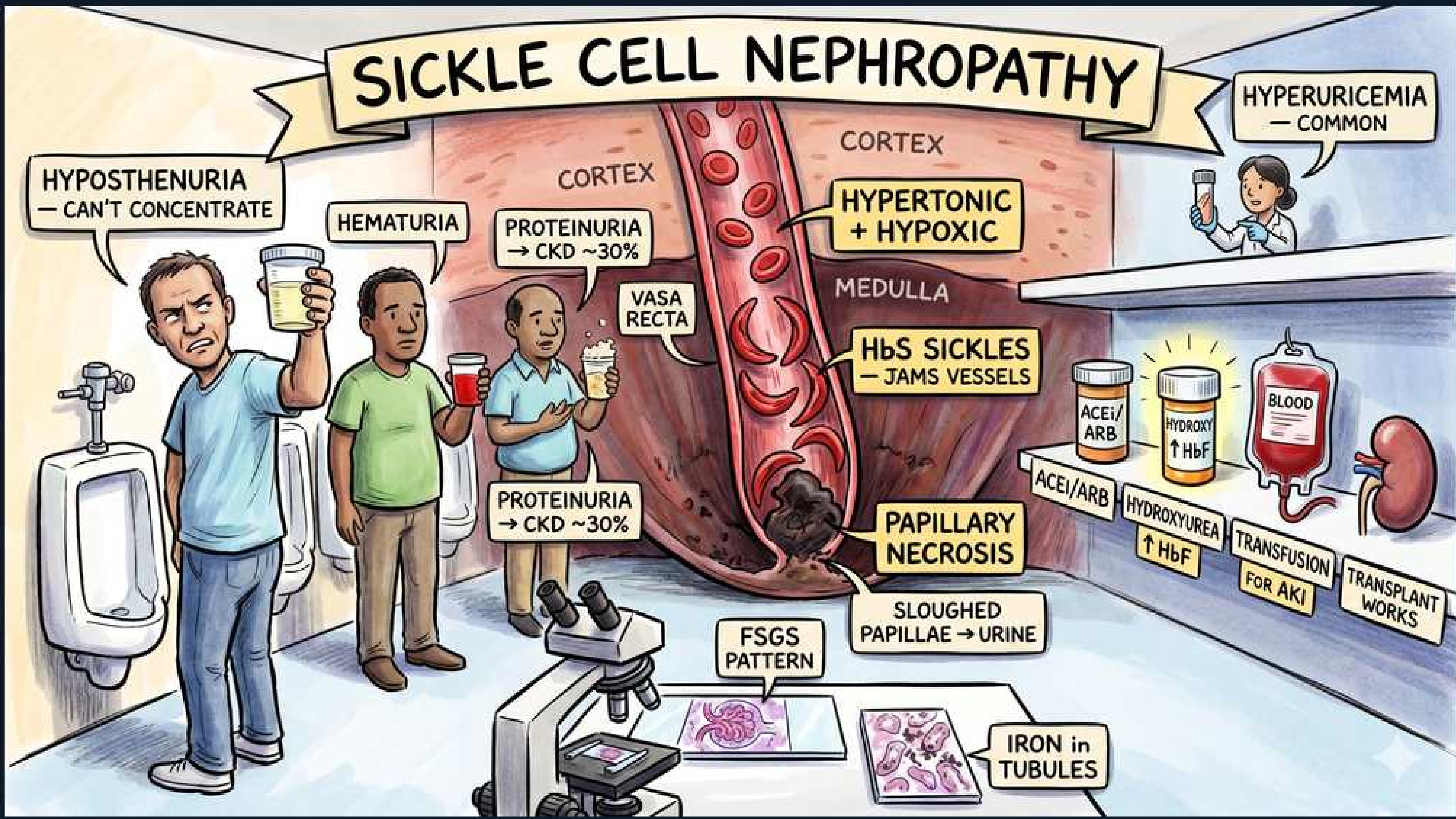
Sickle Cell Nephropathy



1. Hyposthenuria

WHAT YOU SEE

Man at urinal, 'can't concentrate'

WHAT IT ENCODES

Earliest and most universal renal defect in sickle cell disease/trait. Sickling in the hypertonic, hypoxic, acidotic vasa recta of the inner medulla disrupts the countercurrent multiplier, so the kidney loses maximal urine-concentrating ability (isosthenuria). Patients develop nocturia and are prone to dehydration. Reversible early but becomes fixed with repeated medullary injury.

BOARD TRAP

Don't call it diabetes insipidus — DI is ADH-deficiency/resistance, here the defect is medullary architecture from vaso-occlusion despite normal ADH.

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2. Hematuria

WHAT YOU SEE

Man holding cup of red urine

WHAT IT ENCODES

Painless gross or microscopic hematuria, classically from the LEFT kidney, due to medullary capillary congestion and microinfarcts. Most common renal complication of sickle cell trait specifically. Usually self-limited.

BOARD TRAP

In a young Black patient with painless hematuria, sickle trait/papillary necrosis beats glomerulonephritis — look for normal complement and no RBC casts. But always exclude renal medullary carcinoma.

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3. Hypertonic + Hypoxic Medulla

WHAT YOU SEE

Yellow banner over vasa recta

WHAT IT ENCODES

The renal medulla is the perfect storm for HbS polymerization: low O2 tension, high osmolarity, and low pH all promote deoxygenation and sickling. This is why renal pathology in SCD localizes to the medulla (papillary necrosis, concentrating defect) early and the glomerulus (FSGS) later.

BOARD TRAP

This medullary environment — not the cortex — drives early disease; cortical/glomerular FSGS is a later hyperfiltration consequence, not the initiating lesion.

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4. HbS Sickles / Jams Vessels

WHAT YOU SEE

Yellow text on vasa recta

WHAT IT ENCODES

Sickled RBCs occlude the vasa recta, causing medullary ischemia and infarction. This vaso-occlusion is the unifying mechanism behind papillary necrosis, hyposthenuria, and hematuria. Chronic medullary ischemia also blunts erythropoietin and prostaglandin responses.

BOARD TRAP

Vaso-occlusion (mechanical) is the mechanism, not immune-complex deposition — complement is normal and IF is non-specific.

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5. Papillary Necrosis

WHAT YOU SEE

Yellow banner, lower medulla

WHAT IT ENCODES

Sloughing of necrotic renal papillae from vasa recta occlusion; papillae pass into urine ('ring sign' on imaging). Classic SCD complication. Differential includes the mnemonic POSTCARDS: Pyelonephritis, Obstruction, Sickle cell, TB, Cirrhosis/Chronic alcohol, Analgesics (NSAIDs/phenacetin), Renal vein thrombosis, Diabetes, Systemic vasculitis.

BOARD TRAP

Painful hematuria with passage of tissue fragments — think papillary necrosis, not a simple stone; CT urogram shows clubbed calyces.

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6. Sloughed Papillae → Urine

WHAT YOU SEE

Text under papillary necrosis

WHAT IT ENCODES

Necrotic papillae detach and obstruct, causing colicky pain, hematuria, and possible postrenal AKI. On retrograde pyelogram look for the 'ball-on-tee' or 'ring shadow' of a detached papilla. Analgesic nephropathy and SCD are the classic causes.

BOARD TRAP

Tissue in the urine + flank pain mimics a passing stone; the discriminator is the radiographic ring sign and underlying SCD.

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7. Proteinuria / CKD ~30%

WHAT YOU SEE

Text near middle figure

WHAT IT ENCODES

With age, glomerular hyperfiltration (from chronic anemia and hyperfiltration of surviving nephrons) drives albuminuria and progressive CKD in ~30% of homozygous SCD patients. ACE inhibitors/ARBs reduce proteinuria. Sickle nephropathy is a major cause of ESRD in this population.

BOARD TRAP

Proteinuria here is a late hyperfiltration/FSGS phenomenon, distinct from the early medullary defects — don't lump it with the hematuria/concentrating problems.

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8. FSGS Pattern

WHAT YOU SEE

Microscope with glomerulus slide

WHAT IT ENCODES

The dominant glomerular lesion in chronic sickle nephropathy is secondary FSGS, often with glomerulomegaly from chronic hyperfiltration. Presents with nephrotic-range proteinuria in adults. Mechanism is adaptive hyperfiltration, not a primary podocytopathy.

BOARD TRAP

This is SECONDARY FSGS (treat the driver + RAAS blockade), not primary FSGS — high-dose steroids are not the answer.

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9. Iron in Tubules

WHAT YOU SEE
Pink tissue slides, lower right

WHAT IT ENCODES
Chronic intravascular hemolysis releases hemoglobin; tubular cells reabsorb it and accumulate hemosiderin (hemosiderinuria). Stains positive with Prussian blue. Reflects ongoing hemolysis and may contribute to tubular injury.

BOARD TRAP
Iron-laden tubules reflect chronic hemolysis, not a primary nephrotoxin ingestion or rhabdomyolysis.

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10. Hyperuricemia

WHAT YOU SEE
Person, upper right

WHAT IT ENCODES
Increased RBC turnover raises urate production while medullary damage reduces urate handling, causing hyperuricemia and gout in SCD. Common but rarely the board's main point.

BOARD TRAP
Hyperuricemia is a marker of high cell turnover, not evidence of tumor lysis or primary gout.

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11. ACEi/ARB

WHAT YOU SEE
Pill bottle labeled ACEi/ARB

WHAT IT ENCODES
First-line to reduce glomerular hyperfiltration and proteinuria in sickle nephropathy, slowing CKD progression. Monitor potassium and creatinine given medullary dysfunction (type 4 RTA tendency).

BOARD TRAP
RAAS blockade targets the proteinuria/FSGS arm; it does NOT fix the concentrating defect or prevent vaso-occlusive crises.

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12. Hydroxyurea / ↑HbF

WHAT YOU SEE
Blood bag, '↑HbF'

WHAT IT ENCODES
Hydroxyurea raises fetal hemoglobin (HbF), which resists polymerization, reducing sickling, vaso-occlusive crises, and likely slowing nephropathy. Disease-modifying backbone of SCD therapy.

BOARD TRAP
Hydroxyurea works by inducing HbF, not by simple cytoreduction — its renal benefit is from less sickling, not from lowering WBC.

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13. Transfusion for AKI

WHAT YOU SEE
Blood transfusion bag

WHAT IT ENCODES
Exchange/simple transfusion dilutes HbS and improves oxygenation during severe vaso-occlusive renal injury or acute on chronic decline. Lowers HbS percentage to reduce ongoing sickling.

BOARD TRAP
Transfusion is supportive for acute crises; it is not curative and over-transfusion risks iron overload and alloimmunization.

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14. Transplant Works

WHAT YOU SEE
Kidney icon, far right

WHAT IT ENCODES
For ESRD from sickle nephropathy, kidney transplantation has acceptable outcomes; survival is better than long-term dialysis. Sickling can recur in the allograft medulla but graft survival is reasonable.

BOARD TRAP
ESRD here is treatable with transplant — don't assume SCD is an absolute contraindication; outcomes outperform dialysis.
